Michal Balcerak

Kempner Research Fellow | Harvard University

Website: michalbalcerak.ai | Papers: Scholar | Email: m1balcerak@gmail.com
Location: Cambridge, MA, USA

Research Focus: Studying the foundations of generative modeling, optimization-based reasoning, and AI for science through sampling and probabilistic inference, drawing on optimal transport theory and energy-based models.

EDUCATION

PhD, Computer Science 2022 – 2026
University of Zurich & ETH AI Center, Switzerland
Thesis: "Optimization-Based Composable Generation with Learnable and Scientific Priors"

Master of Science, Physics 2020 – 2022
University of Goettingen, Germany
Thesis: "Dynamical System Modeling via Koopman Learning"

Bachelor of Science, Physics 2014 – 2017
Jagiellonian University, Cracow, Poland;
Top 10%, Thesis: "Algebraic Approach to the Supersymmetric Calogero-Sutherland Model"

SELECTED FIRST-AUTHOR PUBLICATIONS

NeurIPS (2026)	"Graph Energy Matching: Transport-Aligned Energy-Based Modeling for Graph Generation" [paper] [code] [project page]
NeurIPS (2025)	"Energy Matching: Unifying Flow Matching and Energy-Based Models for Generative Modeling" [paper] [code, 225+ GitHub stars]
Nature Communications (2025)	"Individualizing Glioma Radiotherapy Planning by Optimization of a Data- and Physics-Informed Discrete Loss" [paper] [code]
NeurIPS (2024)	"Physics-Regularized Multi-Modal Image Assimilation for Brain Tumor Localization" [paper] [code]

10 publications in total (first- and co-authored), incl. IEEE TMI and Medical Image Analysis.

SELECTED TALKS

Featured Speaker & Closing Panelist, Frontiers of Flows for Generative AI, Mathematical Foundations of AI Workshop, Carnegie Mellon University, Pittsburgh, PA [event website] 03/2026

Featured Speaker, Inverse Biophysical Modeling and Machine Learning in Personalized Oncology, Dagstuhl Seminar 23022, Schloss Dagstuhl, Germany [event website] 01/2023

EXPERIENCE

Kempner Research Fellow 2026 – present
Kempner Institute, Harvard University, Cambridge, MA, USA
- Research on optimization-based reasoning and generative modeling.

Graduate Researcher 2022 – 2026
University of Zurich / ETH AI Center, Zurich, Switzerland
- Developed optimization-based generative modeling and scientific inverse-problem methods using energy-based models, optimal transport, and physics-informed priors.
- Led model development, evaluation, and open-source releases.

Visiting Research Fellow 2023 – 2024
School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA
- Developed hybrid neural networks and classical methods to estimate cancer cell locations on discrete, graph-like representations of brain soft tissues, incorporating biomechanical elasticity and PDE-based priors in Prof. Petros Koumoutsakos' lab. Supported by a GRC Travel Grant for a six-month research stay.

Visiting Research Fellow 2023
Technical University of Munich, Munich, Germany
- Combined patient imaging with tumor-growth and tissue-elasticity priors to estimate tumor cell distributions during a research visit with Profs. Barbara Wohlmuth and Benedikt Wiestler.

Founder & CTO 2018 – 2020
Acai Analytics sp. z o.o., Cracow, Poland
- Led a small team of data scientists and software engineers developing machine-learning trading strategies with MCMC-based forecasting and risk analysis.

Student Researcher 2015 – 2018
CERN, Geneva, Switzerland
- Developed a machine learning framework to identify collision topologies at the Large Hadron Collider within the ATLAS experiment, focusing on anomaly detection. Worked on-site for four months and continued the project remotely thereafter.

SKILLS

Languages:	English (C2), German (B1), Polish (Native)
Programming:	Python, C++, JAX, PyTorch, TeX/Tikz
Systems:	Linux, Git, Docker, GPU clusters/HPC, cloud computing
Review:	NeurIPS, ICLR, ICML, Medical Image Analysis

AWARDS

Kempner Research Fellowship (2026): Postdoctoral fellowship awarded by the Kempner Institute for the Study of Natural and Artificial Intelligence, Harvard University.

GRC Travel Grant (2023): Awarded by Graduate Campus, University of Zurich, for a research stay at Harvard University.

DAAD Study Scholarship (2019): Awarded to 15 out of 80 applicants, covering full MSc expenses in Germany.

Minister of Science and Higher Education Scholarship (2015, 2016): Awarded annually to top ∼500 students in Poland.

REFERENCES

- **Prof. Dr. Bjoern Menze**, bjoern.menze@uzh.ch
  University of Zurich & ETH AI Center
  (Main PhD Supervisor, Co-author)
- **Prof. Dr. Yilun Du**, ydu@seas.harvard.edu
  Kempner Institute, Harvard University
  (Research Co-supervisor, Co-author)
- **Prof. Dr. Benedikt Wiestler**, b.wiestler@tum.de
  Technical University of Munich
  (PhD Co-supervisor, Co-author)